

PIA PROJETO

TECNOLOGIAS E SISTEMAS DE INFORMAÇÃO PARA A WEB

PROJETO

Subject

Employee: Modelagem Preditiva com Machine Learning

Objectives and brief description

1. Introduction

In a modern web systems architecture, AI models do not operate in isolation and rely on a rigorous chain of data transformations that must be perfectly reproducible outside of the original development environment.

This project aims to consolidate the knowledge acquired in the Artificial Intelligence Curricular Unit, challenging students to solve a real end-to-end predictive modeling problem.

Each working group should build two independent predictive models applied on the same dataset:

Regression Model: Predict the monthly salary (`MonthlyIncome`).

Classification Model: Predict whether the employee will resign (`Attrition`).

2. The Dynamics of Evaluation (Blind Test)

The evaluation of this project will be based on a Blind Test on the day of presentation and defense. At that time, the group models will be fed with data through a script that will get a prediction and generate a ranking leaderboard in real time.

3. Technical Requirements and the Pipeline Architecture

To ensure full portability of the solution created, groups should not export the models and data transformations in separate files. Instead, the use of the Scikit-Learn Pipeline class is mandatory.

The Pipeline will encapsulate the entire engineering flow: from imputation, encoding, scaling, variable selection, to the final algorithm. In this way, the exported file will act as a "black box" capable of receiving raw data in the original format and producing forecasts without manual intervention.

Files to be delivered (via Moodle):

[GroupX]_pipeline_regression.pkl: The complete pipeline trained for the regression problem.

[GroupX]_pipeline_classification.pkl: The complete pipeline trained for the classification problem.

[GroupX]_notebook.ipynb: The detailed Jupyter Notebook that gave rise to the models (containing the Exploratory Analysis, algorithm validations, hyperparameter tuning, and theoretical commentary).

[GroupX]_report.docx: The detailed report on project execution, decisions taken, and the reasoning behind them, models tested and GenAI queries used.

4. Mandatory Code Structure for Export

Files should be serialized using Python's native pickle library.

Groups should validate their solution with the test dataset.

5. Evaluation Criteria (Total: 20 Points)

Quality of Data Engineering and Pipelines (3 Points): Robustness of the pipeline in the autonomous treatment of missing data, appropriate encoding strategies (e.g. OneHotEncoder dealing with unknown categories) and correct selection of features.

Exploration and Scientific Methodology (4 Values): Depth of Exploratory Data Analysis (EDA) evidenced in the Notebook, comparative experimentation of different algorithms and correct use of hyperparameter optimization techniques (e.g., GridSearchCV or RandomizedSearchCV).

Blind Test Performance Metrics (6 Points): Absolute performance and relative placement on the official Leaderboard. The evaluation metrics will be the R2/RMSE for the regression and the F1-Score for the classification.

Organization of Code and Documentation (3 Points): Strict compliance with the rules of file naming, readability and clean structure of the delivered Notebook and report.

Face-to-face Oral Defense (4 Points): Responsiveness and theoretical justification of the group members regarding the engineering decisions and algorithms adopted during the presentation session.

6. Final Notes and Deadlines

The use of Generative AI tools is valid as a code optimization assistant, however students should exercise full analytical control over the project, the hyperparameters and the robustness of the cross-validation. Queries used in interaction with Generative AI should be included in the report.

The submission of the three files described in Section 3 has a deadline of 11:59 pm on June 18, 2026. Undelivered or corrupted files will not be admitted.

The presentation and defense will take place, for all groups, on June 19 from 9:00 am.